

Samuel Koovely

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EXPERIENCE	<i>Teaching Assistant</i>	Sep. 2022 - Ongoing
	University of Zurich, Switzerland	
	• Organizer of <i>Introduction to Machine Learning</i>	Spring Term 2025
	• Organizer of <i>Markov Models</i>	Spring Term 2024
	• (Head) Assistant of <i>Complex Networks Theory and Applications</i>	Fall Terms 2022-2025
	<i>Project Intern and Trainee</i>	Nov. 2020 - Nov. 2021
	IBM Research Zurich, Switzerland	
	• Project intern for MSc Thesis [5]	
	• Further developed the model introduced in [5] as a post-graduate trainee.	
	<i>Teaching Assistant</i>	Sep. 2017 - Aug. 2019
	ETH Zurich, Switzerland	
	• Assistant of <i>Mathematik IV: Statistik</i> (German)	Spring Term 2019
EDUCATION	• Assistant of <i>Stochastics</i> (German)	Fall Term 2018
	• Assistant of <i>Probability and Statistics</i>	Fall Term 2017
	<i>PhD studies in Mathematics</i>	Aug. 2022 - ongoing
	University of Zurich, Institute of Mathematics, Institute of Mathematical Modeling and Machine Learning	
	<i>BSc and MSc studies in Mathematics</i>	Sep. 2015 - Jul. 2021
	ETH Zurich, Department of Mathematics	
SKILLS	Languages: Italian (native), English (proficient), French (good), German (good) Programming Skills: Python (good), R (basic), Matlab (basic), C++ (basic), Wolfram Language (basic)	
TALKS AT WORKSHOPS & CONFERENCES	<i>Evolution of Conditional Entropy for Diffusion Dynamics on Graphs</i> , NetSci 2026	Jun. 2026
	<i>Entropy-Informed Structural Changes Detection</i> , NetSci 2025	Jun. 2025
	<i>What is... Free Convolution?</i> , ZGSM Graduate Colloquium, ETH/UZH	Mai 2025
	<i>Entropy-Informed Structural Changes Detection</i> , Workshop: Temporal and Dynamic Interactions, Oxford	February 2025
	<i>Information-theoretic framework for the dynamics of temporal networks</i> , NetSci 2024	Jun. 2024
	<i>Random Walks and Community Detection</i> , Complex Networks: Theory, Methods, and Applications - 7th ed.	May 2023

SERVICES TO COMMUNITY	<i>Oragnization:</i> ZGSM Graduate Colloquium, ETH/UZH	Jul. 2025-ongoing
	<i>Reviewer for Journals:</i> Physical Review E (2025)	
	<i>Reviewer for Conferences:</i> NetSci (2025)	
	<i>Examiner of candidates high-school teacher in mathematics,</i> Kantonsschule Rämibühl MNG	Oct. 2024
	<i>Committee member for professorial appointments, UZH</i>	2023-2025

PREPRINTS & PUBLICATIONS	[1]	Samuel Koovely and Alexandre Bovet. “Conditional Entropy of Heat Diffusion on Temporal Networks”. In: https://arxiv.org/pdf/2605.21514 (2026).
	[2]	Samuel Koovely. “Generating Temporal Networks with the Ascona Model”. In: https://arxiv.org/pdf/2512.16972 (2025).
	[3]	Samuel Koovely and Alexandre Bovet. “Evolution of Conditional Entropy for Diffusion Dynamics on Graphs”. In: https://arxiv.org/pdf/2510.19441 (2025).

THESES & PROJECTS	[4]	Samuel Koovely. “Theory and Applications of Conditional Entropy of Heat Diffusion on Temporal Networks”. PhD Thesis. 2026.
	[5]	Samuel Koovely. “A mathematical framework for COMIC-Tree: an undirected graphical model for T-cell receptors specificity”. MSc Thesis. 2021.
	[6]	Samuel Koovely. “Overview and empirical evaluation of some variations of the PC-algorithm”. Semester Research Project. 2020.
	[7]	Samuel Koovely. “Introduction to Riemann surfaces and covering spaces”. BSc Thesis. 2019.